Homework 3.2: Benamou-Brenier in One Dimension

Introduction

This homework studies the dynamic optimal transport interpretation of the Benamou-Brenier formula in one dimension.

Given two probability densities

$$\rho_0$$

and

$$\rho_1,$$

with cumulative distribution functions

$$F_0$$

and

$$F_1,$$

the monotone one-dimensional optimal transport map is

$$T(x) = F_1^{-1}(F_0(x)).$$

The corresponding Wasserstein displacement interpolation is

$$X_t(x) = (1-t)x + tT(x), \qquad 0 \le t \le 1,$$

and the intermediate probability density is

$$\rho_t = (X_t)_{\#}\rho_0.$$

The goals of the experiment were to:

- compute and verify the one-dimensional optimal transport map;
- visualize the displacement interpolation;
- estimate the squared Wasserstein distance;
- numerically evaluate the Benamou-Brenier action;
- compare compactly supported and Gaussian target distributions;
- evaluate the mathematical reliability of LLM-generated reasoning and code.

Part I: LLM Plan and Preliminary Verification

Prompt Given to the LLM

> Given two one-dimensional probability densities ρ_0 and ρ_1, how can I compute the optimal transport map using CDFs and quantile functions? How can I use this map to animate the Benamou-Brenier displacement interpolation?

LLM Plan

The LLM proposed computing

$$F_0(x) = \int_{-\infty}^{x} \rho_0(s) \, ds$$

and

$$F_1(x) = \int_{-\infty}^{x} \rho_1(s) \, ds.$$

The optimal transport map is then

$$T(x) = F_1^{-1}(F_0(x)).$$

Once the map is known, particles move according to

$$X_t(x) = (1-t)x + tT(x).$$

The density is transported through

$$\rho_t = (X_t)_{\#}\rho_0.$$

The proposed numerical procedure was:

1. evaluate the source and target densities on a grid;
2. normalize the densities;
3. compute their CDFs;
4. construct the target quantile function;
5. calculate $T(x)$;

6. calculate $X_t(x)$;

7. reconstruct the intermediate densities;

8. plot the resulting transport.

Preliminary Sanity Check

Before applying the method to the homework densities, I tested the implementation on two Gaussian distributions:

$$\rho_0 \approx N(-2, 1)$$

and

$$\rho_1 \approx N(2, 1).$$

Since the target is exactly a translation of the source by four units, the exact optimal map is

$$T(x) = x + 4.$$

The numerical calculation gave an average displacement

$$3.9999999642,$$

with maximum interior error approximately

$$2.93 \times 10^{-7}.$$

The defining quantile identity

$$F_1(T(x)) = F_0(x)$$

had maximum numerical error

$$1.11 \times 10^{-16}.$$

The squared Wasserstein distance should be

$$W_2^2 = 16,$$

and the numerical calculation gave

$$W_2^2 = 15.9999999028.$$

Thus the CDF, quantile, and displacement-interpolation implementation was verified before being used for the main experiment.

Part II: Half-Circle Density to Uniform

Density

The initial density is

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1,$$

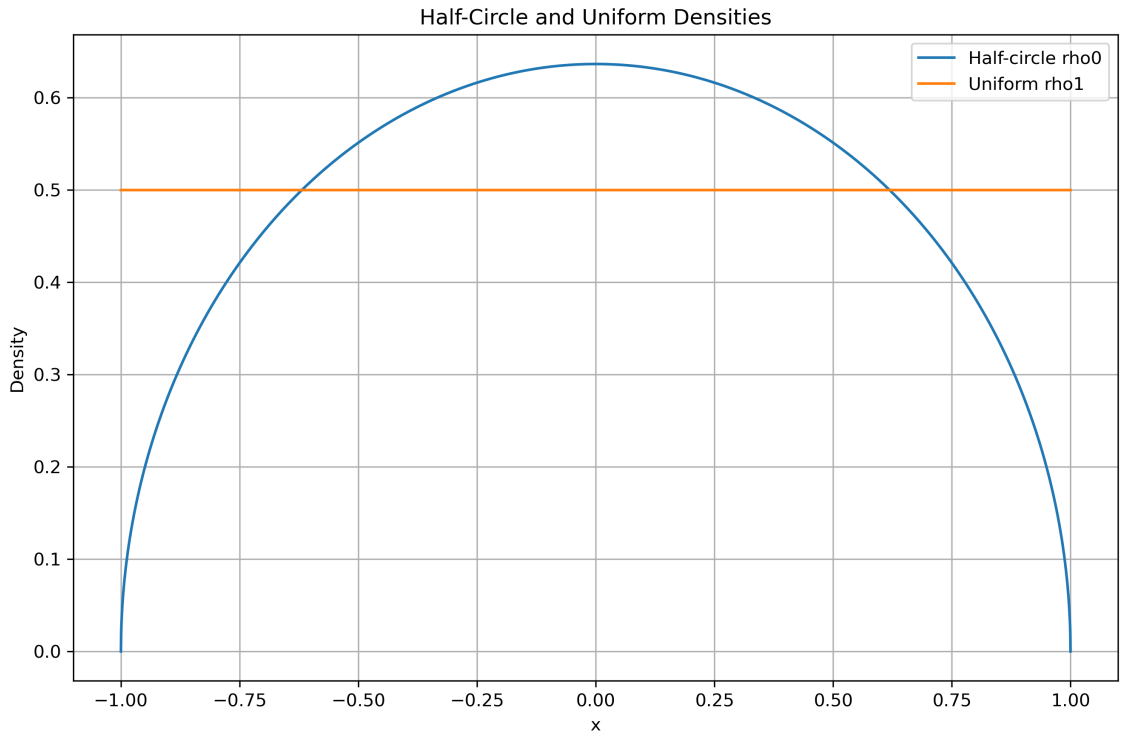
and zero outside $[-1, 1]$.

The target density is uniform:

$$\rho_1(x) = \frac{1}{2}, \qquad -1 \le x \le 1,$$

and zero outside $[-1, 1]$.

Source and Target Densities



The half-circle density is concentrated more strongly around the origin, whereas the uniform target distributes the probability mass evenly across the interval.

CDFs

For the source,

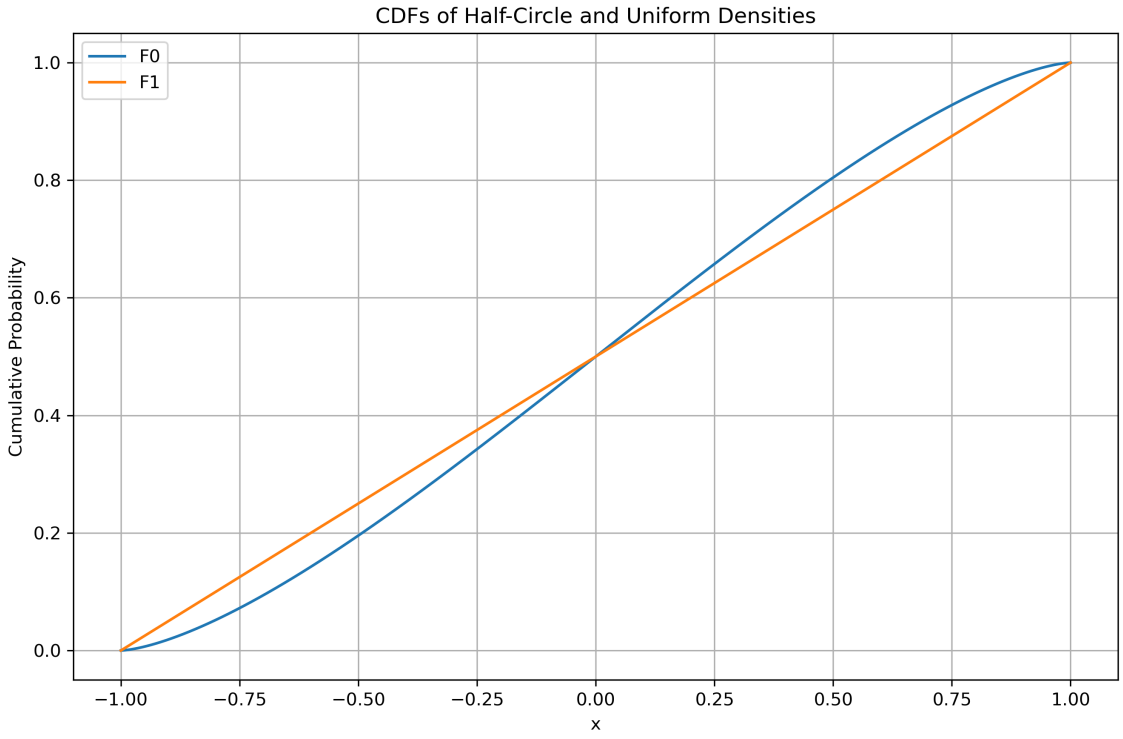
$$F_0(x) = \frac{1}{2} + \frac{1}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right), \qquad -1 \le x \le 1.$$

For the target,

$$F_1(x) = \frac{x+1}{2}, \qquad -1 \le x \le 1.$$

Thus

$$F_1^{-1}(u) = 2u - 1.$$



The numerical and analytic CDFs agreed closely.

The maximum error for the numerical source CDF was approximately

$$2.03 \times 10^{-6}.$$

Optimal Transport Map

The optimal map is

$$T(x) = F_1^{-1}(F_0(x)).$$

Therefore,

$$T(x) = 2F_0(x) - 1,$$

which gives

$$T(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The map satisfies

$$T(-1) = -1,$$

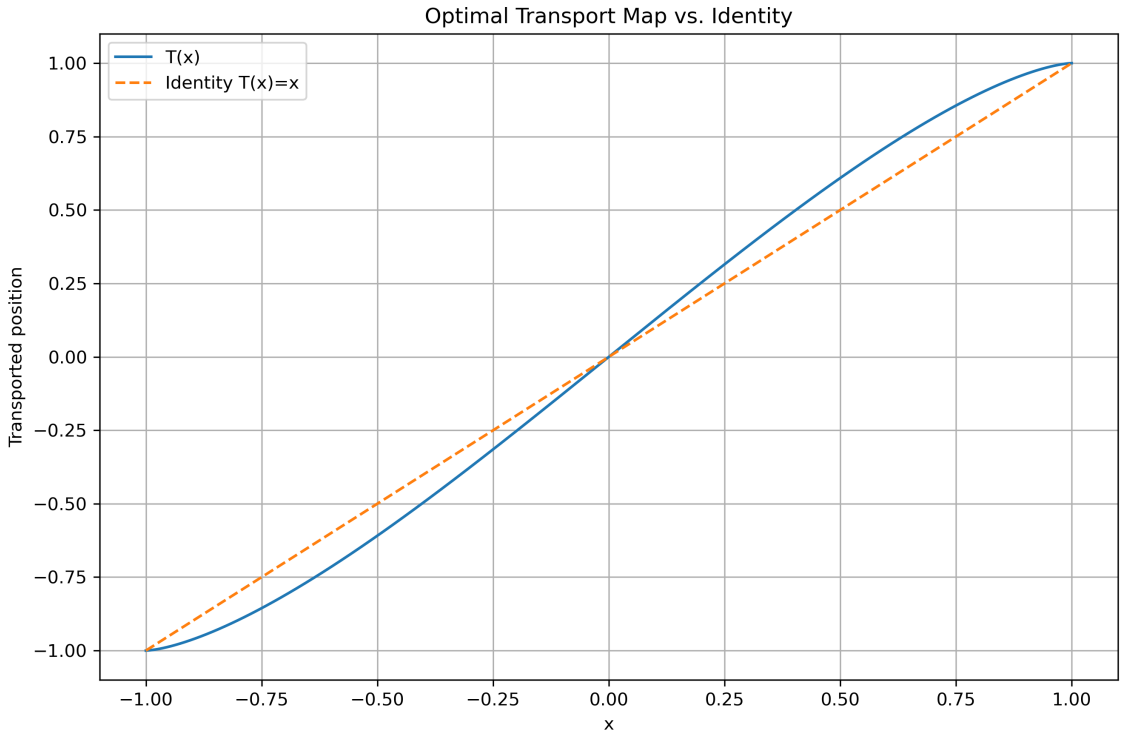
$$T(0) = 0,$$

and

$$T(1) = 1.$$

The numerical and analytic transport maps differed by at most approximately

$$4.05 \times 10^{-6}.$$



For

$$x > 0,$$

we have

$$T(x) > x,$$

so particles move outward to the right.

For

$$x < 0,$$

we have

$$T(x) < x,$$

so particles move outward to the left.

This is exactly what is required to flatten the initially peaked density.

Displacement Interpolation

The Wasserstein geodesic is

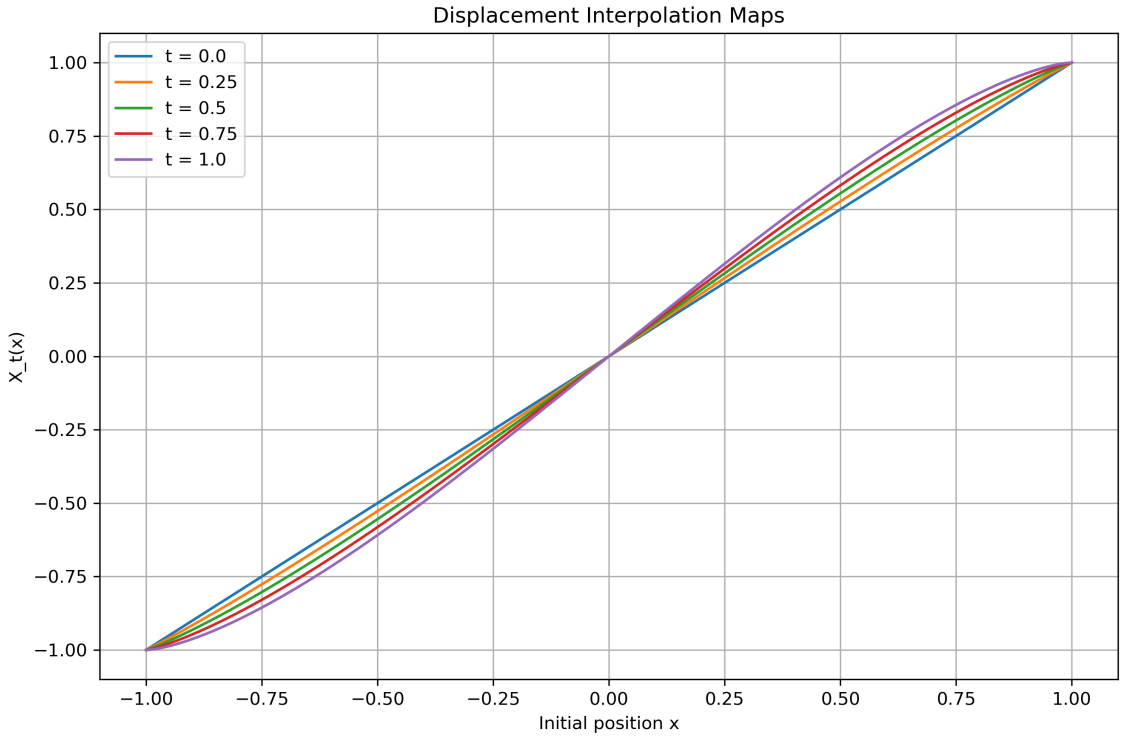
$$X_t(x) = (1-t)x + tT(x).$$

Substituting the transport map,

$$X_t(x) = (1-t)x + \frac{2t}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The maps were evaluated at

$$t = 0,\ 0.25,\ 0.5,\ 0.75,\ 1.$$



The interpolation begins with the identity map and continuously approaches the optimal transport map.

Interpolated Densities

Differentiating the transport map gives

$$T'(x) = \frac{4}{\pi}\sqrt{1-x^2}.$$

Therefore,

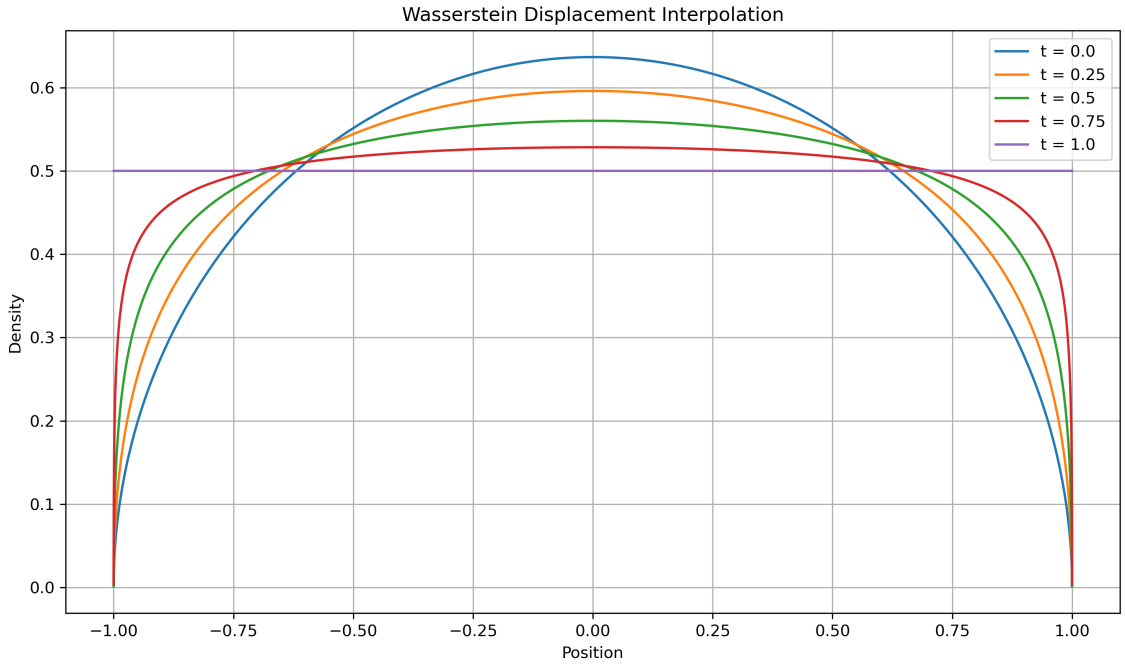
$$X_t'(x) = (1-t) + \frac{4t}{\pi}\sqrt{1-x^2}.$$

Conservation of mass gives

$$\rho_t(X_t(x))X_t'(x) = \rho_0(x).$$

Hence,

$$\rho_t(X_t(x)) = \frac{\frac{2}{\pi}\sqrt{1-x^2}}{(1-t) + \frac{4t}{\pi}\sqrt{1-x^2}}.$$



At

$$t = 0,$$

the density is the half-circle.

As t increases, the central peak decreases and probability mass moves outward.

At

$$t = 1,$$

the density becomes

$$\rho_1(x) = \frac{1}{2}.$$

The total mass remained approximately one for every tested time.

Velocity Along Particle Paths

The particle trajectory is

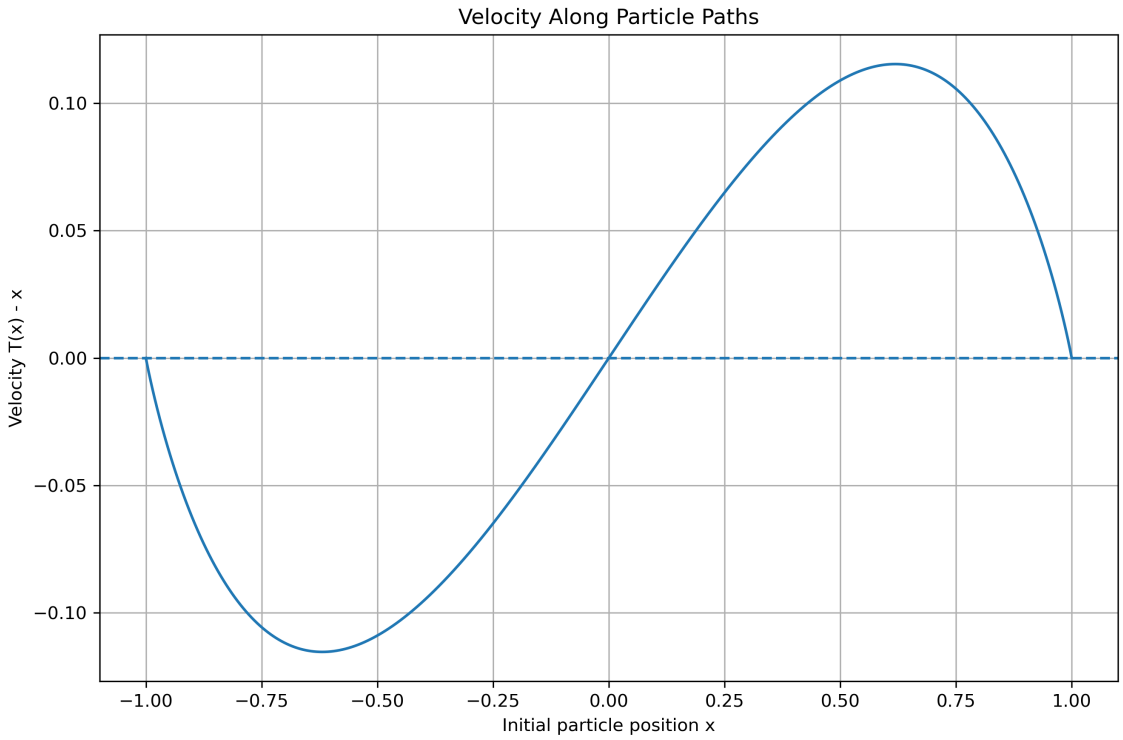
$$X_t(x) = (1-t)x + tT(x).$$

Differentiating with respect to time gives

$$\dot{X}_t(x) = T(x) - x.$$

Therefore,

$$\dot{X}_t(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right) - x.$$



The velocity is negative on the left side and positive on the right side.

Thus particles move outward from the center.

The velocity is zero at

$$x = -1, \quad 0, \quad 1.$$

Each particle has constant **Lagrangian** velocity along its own displacement-interpolation path. The corresponding Eulerian velocity field is

$$v_t(y) = T(x) - x \quad \text{where } y = X_t(x),$$

so at a fixed spatial location y the velocity field can depend on time through the inverse map $x = X_t^{-1}(y)$.

Part III: Numerical Wasserstein Distance and Benamou-Brenier Action

The squared Wasserstein distance is

$$W_2^2(\rho_0, \rho_1) = \int_{-1}^{1} |T(x) - x|^2 \rho_0(x) \, dx.$$

For this problem,

$$W_2^2 = \int_{-1}^{1} \left[\frac{2}{\pi} \left(\arcsin x + x\sqrt{1-x^2} \right) - x \right]^2 \frac{2}{\pi} \sqrt{1-x^2} \, dx.$$

The numerical estimate was approximately

$$W_2^2 \approx 0.00692838.$$

As an additional check, the analytic benchmark is

$$W_2^2 = \frac{7}{12} - \frac{256}{45\pi^2},$$

which is also approximately

$$0.00692838.$$

Benamou-Brenier Action

The dynamic action is

$$\mathcal{A} = \int_0^1 \int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y) \, dy \, dt.$$

Along the particle path,

$$v_t(X_t(x)) = T(x) - x.$$

Using

$$y = X_t(x)$$

and conservation of probability mass,

$$\rho_t(X_t(x))X_t'(x) = \rho_0(x),$$

we obtain

$$\int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y) \, dy = \int_{-1}^{1} |T(x) - x|^2 \rho_0(x) \, dx.$$

Therefore,

$$A(t) = W_2^2$$

for every t theoretically.

Numerically, the instantaneous action remained approximately

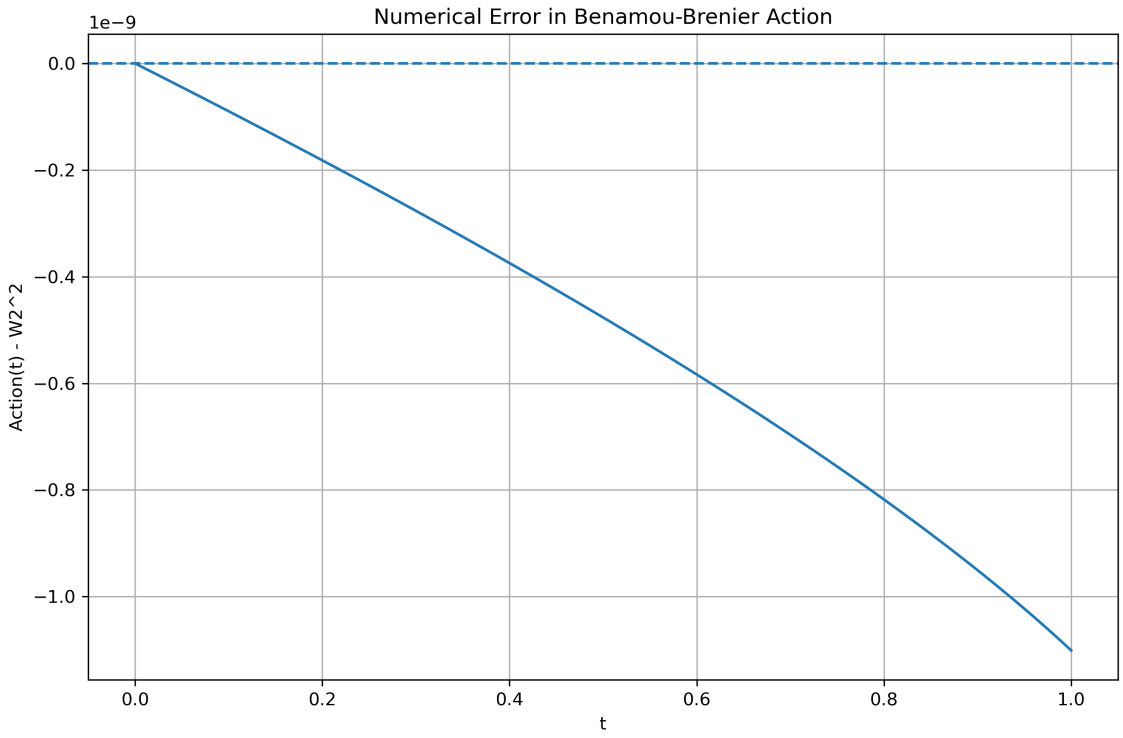
$$6.92837 \times 10^{-3}.$$

The difference

$$A(t) - W_2^2$$

remained only on the scale of

$$10^{-9}.$$



At the largest observed discrepancy, the relative error was only on the order of

$$10^{-7}.$$

Therefore, the numerical dynamic action satisfies

$$\mathcal{A} \approx 0.00692838$$

and agrees with

$$W_2^2 \approx 0.00692838.$$

Thus,

$$\mathcal{A} \approx W_2^2$$

to very high numerical accuracy.

This shows that the numerical implementation is consistent with the Benamou-Brenier identity to high accuracy. The equality itself is an analytical theorem; the computation is a verification of the discretization and implementation.

Part IV: Compact Support to Gaussian

The source remains

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1.$$

The target is now the standard Gaussian:

$$\rho_1(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

Let

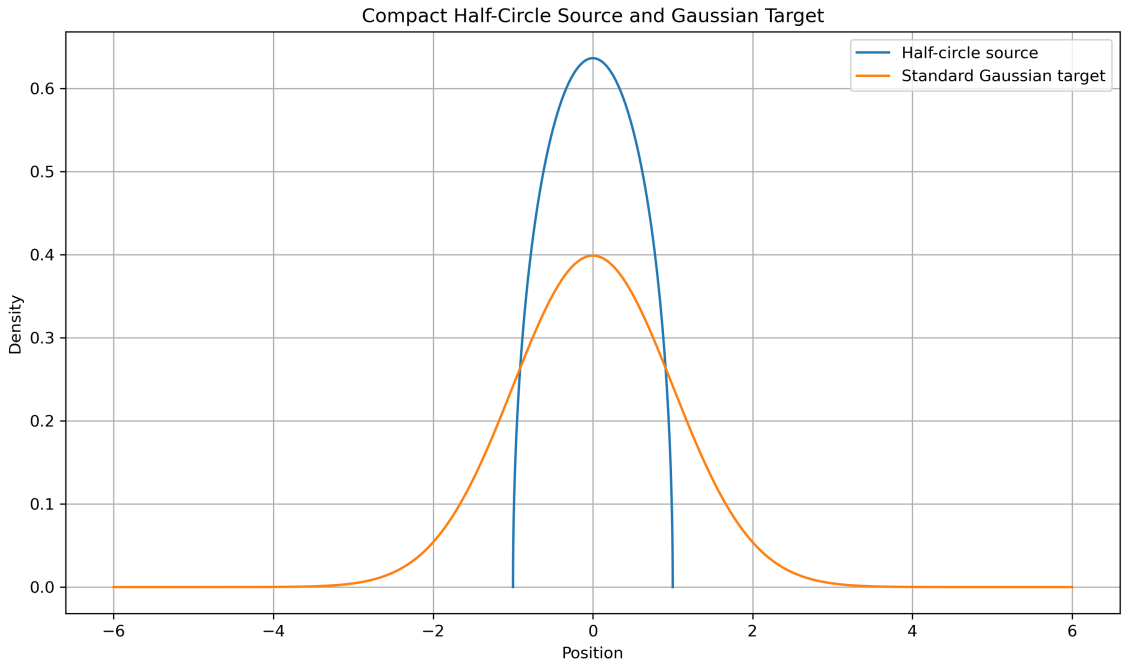
$$\Phi(y)$$

be its CDF.

The transport map is

$$T(x) = \Phi^{-1}(F_0(x)).$$

Compact Source and Gaussian Target



The source has support

$$[-1, 1],$$

whereas the Gaussian has support

$$\mathbb{R}.$$

This difference produces a major change in the optimal transport map.

Gaussian Transport Map

Since

$$F_0(-1) = 0$$

and

$$F_0(1) = 1,$$

we have

$$T(-1) = \Phi^{-1}(0) = -\infty,$$

and

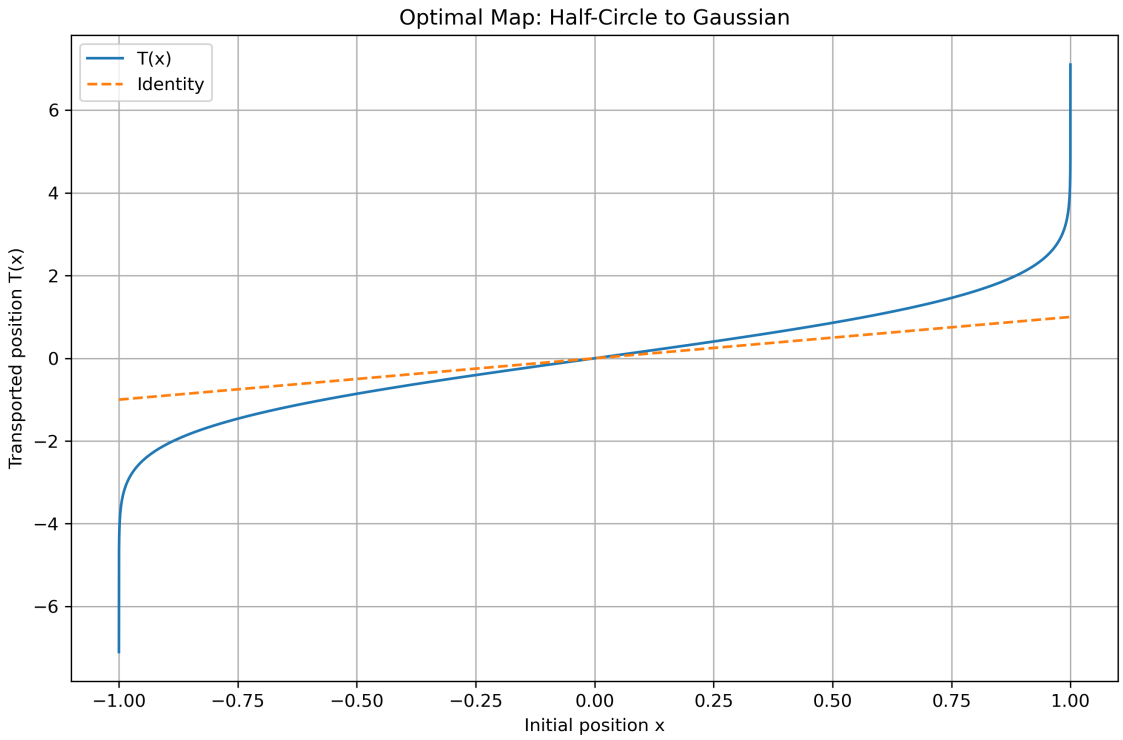
$$T(1) = \Phi^{-1}(1) = +\infty.$$

Therefore,

$$T(x) \to -\infty \quad \text{as } x \to -1^+$$

and

$$T(x) \to +\infty \quad \text{as } x \to 1^-.$$



The numerical endpoint experiment gave:

δ	$T(-1+\delta)$	$T(1-\delta)$
10^{-1}	-2.0815	2.0815
10^{-2}	-3.2392	3.2392
10^{-3}	-4.1196	4.1196

δ	$T(-1+\delta)$	$T(1-\delta)$
10^{-4}	-4.8556	4.8556
10^{-6}	-6.0802	6.0802
10^{-8}	-7.1025	7.1025

This clearly demonstrates the divergence of the transport map.

Interpolated Gaussian Densities

Differentiating

$$\Phi(T(x)) = F_0(x)$$

gives

$$\phi(T(x))T'(x) = \rho_0(x),$$

where

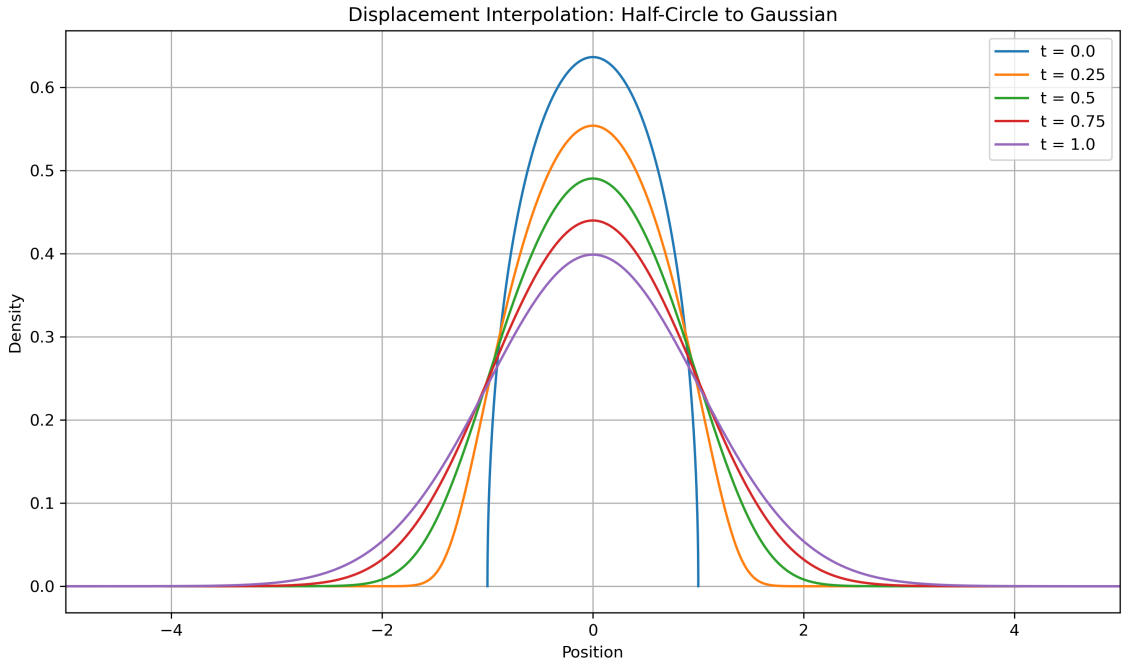
$$\phi(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2}.$$

Hence,

$$T'(x) = \frac{\rho_0(x)}{\phi(T(x))}.$$

The intermediate density satisfies

$$\rho_t(X_t(x)) = \frac{\rho_0(x)}{(1-t) + tT'(x)}.$$



The compact half-circle density gradually spreads into distributions with longer tails.

At

$$t = 1,$$

the resulting density is the standard Gaussian.

The maximum final density error compared with the exact Gaussian was approximately

$$5.55 \times 10^{-17}.$$

Probability mass was also preserved to approximately

$$10^{-6}.$$

Why the Mass Spreads to All of $\mathbb{R}$

Initially,

$$\mathrm{supp}(\rho_0) = [-1, 1].$$

The Gaussian satisfies

$$\rho_1(y) > 0$$

for every finite y.

Thus,

$$\mathrm{supp}(\rho_1) = \mathbb{R}.$$

The quantile map sends probability level u to

$$\Phi^{-1}(u).$$

Since

$$\Phi^{-1}(u) \to -\infty$$

as

$$u \to 0,$$

and

$$\Phi^{-1}(u) \to +\infty$$

as

$$u \to 1,$$

source particles arbitrarily close to -1 and 1 must move arbitrarily far into the tails.

Furthermore, for every fixed

$$t > 0,$$

$$X_t(x) = (1-t)x + tT(x).$$

Since $T(x)$ diverges at the source endpoints,

$$X_t(x)$$

also has unbounded range for every positive time.

Thus the Wasserstein geodesic develops arbitrarily small tails across all of $\mathbb{R}$ immediately after $t = 0$.

Gaussian Truncation

A computer cannot represent the entire real line, so the Gaussian can be approximated on

$$[-L, L].$$

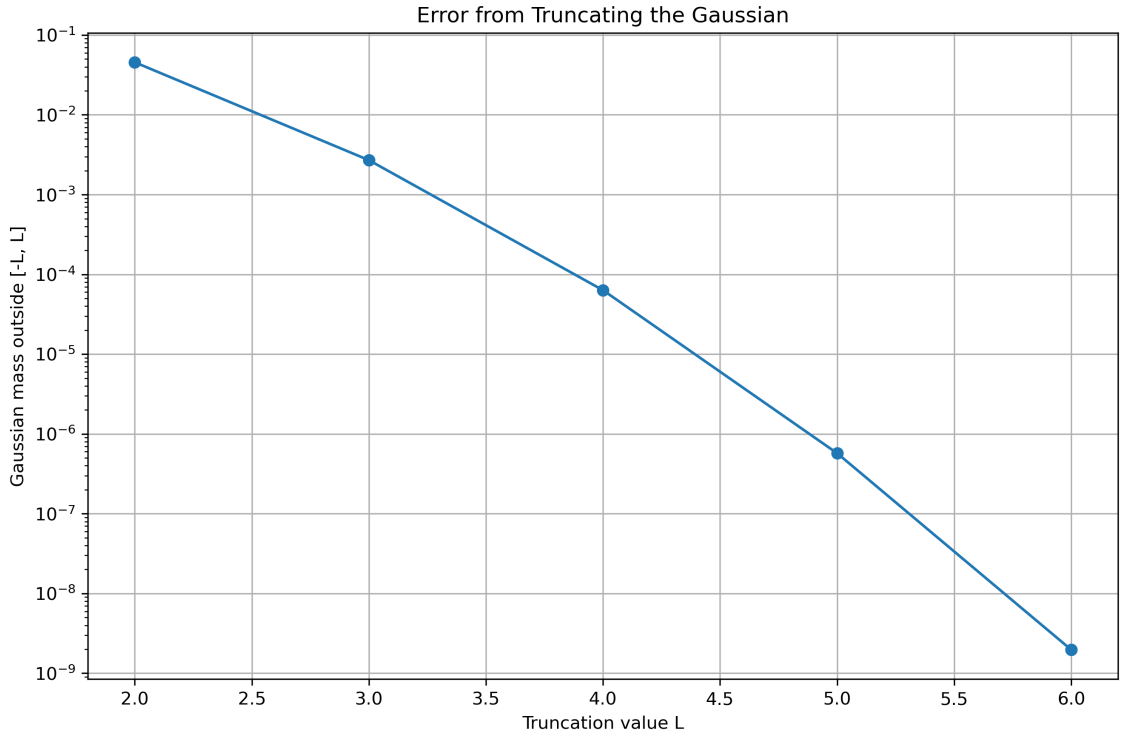
The omitted tail probability is

$$P(|Z| > L) = 2(1 - \Phi(L)).$$

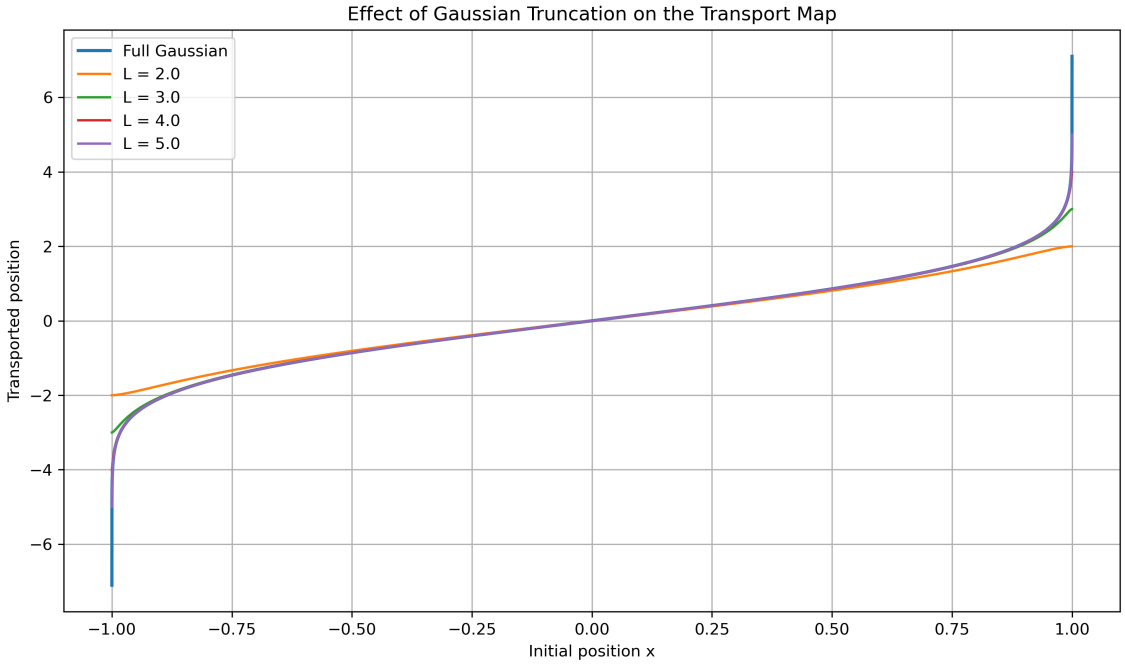
The numerical results were:

L	**Gaussian mass outside** $[-L, L]$
2	4.55×10^{-2}

L	**Gaussian mass outside** $[-L, L]$
3	2.70×10^{-3}
4	6.33×10^{-5}
5	5.73×10^{-7}
6	1.97×10^{-9}



The effect on the transport map is shown below.



For

$$L = 2,$$

the approximation is visibly poor.

At

$$L = 3,$$

the approximation is substantially better.

At

$$L = 4,$$

the omitted mass is only about

$$6.33 \times 10^{-5},$$

so the approximation is visually very good.

At

$$L = 5,$$

only

$$5.73 \times 10^{-7}$$

of the Gaussian probability is omitted.

Therefore,

$$L \approx 4$$

is a reasonable choice for visualization, while

$$L \approx 5$$

provides a very accurate finite-domain approximation.

Part V: LLM Prompts and Critique

Prompt 1: Ask the LLM for a Mathematical Plan

The first prompt given to the LLM was:

> Given two one-dimensional probability densities ρ_0 and ρ_1, how can I compute the optimal transport map using CDFs and quantile functions? How can I use this map to animate the Benamou-Brenier displacement

> interpolation?

The LLM proposed computing the cumulative distribution functions

$$F_0(x) = \int_{-\infty}^{x} \rho_0(s) \, ds$$

and

$$F_1(x) = \int_{-\infty}^{x} \rho_1(s) \, ds,$$

then constructing the one-dimensional monotone transport map

$$T(x) = F_1^{-1}(F_0(x)).$$

It then proposed the displacement interpolation

$$X_t(x) = (1-t)x + tT(x)$$

and the pushforward density

$$\rho_t = (X_t)_{\#}\rho_0.$$

This mathematical plan was correct.

However, I did not simply assume that the LLM's formulas or numerical implementation were correct. Before using the method on the homework distributions, I tested it on a case with a known exact answer.

Using

$$\rho_0 \approx N(-2, 1)$$

and

$$\rho_1 \approx N(2, 1),$$

the exact transport map is

$$T(x) = x + 4.$$

The numerical implementation produced an average displacement of approximately

$$3.9999999642$$

with maximum interior error approximately

$$2.93 \times 10^{-7}.$$

The identity

$$F_1(T(x)) = F_0(x)$$

held with maximum numerical error approximately

$$1.11 \times 10^{-16}.$$

This provided an independent verification that the CDF and quantile procedure proposed by the LLM was being implemented correctly.

Prompt 2: Generate Python Code for the Experiment

I then asked the LLM to generate standalone Python code for the half-circle to Gaussian optimal transport experiment.

The generated code was inspected according to the following questions:

1. Does it normalize the densities correctly?
2. Does it compute the CDFs monotonically?
3. Does it use a generalized inverse or interpolation for the quantile function?
4. Does it handle compact support correctly?
5. Does it handle Gaussian tails honestly rather than pretending infinity fits on a finite grid?

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1. Density Normalization

The source density was

$$\rho_0(x) = \frac{2}{\pi}\sqrt{1-x^2}, \qquad -1 \le x \le 1.$$

Analytically,

$$\int_{-1}^{1} \rho_0(x) \, dx = 1.$$

The numerical integral produced by the LLM-generated code was

$$0.999995814964167.$$

The error was therefore approximately

$$4.19 \times 10^{-6}.$$

This is sufficiently small and is consistent with finite-grid quadrature error.

The Gaussian target required more care because its support is all of

$$\mathbb{R}.$$

The code used

$$[-6, 6]$$

only as a numerical plotting and interpolation interval.

It calculated the Gaussian mass captured inside this interval as

$$0.9999999980268233$$

and explicitly reported that the omitted tail mass was

$$1.97318 \times 10^{-9}.$$

This is mathematically preferable to renormalizing the Gaussian on $[-6, 6]$ and pretending that the finite interval contains the entire distribution.

Therefore, the density normalization was handled correctly.

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2. Monotonicity of the CDFs

The source CDF was computed numerically using cumulative trapezoidal integration.

The result satisfied

$$F_0(-1) = 0$$

and

$$F_0(1) = 1.$$

The program explicitly checked that the numerical CDF was nondecreasing, and the result was

$$\mathrm{True}.$$

The numerical source CDF was also compared with the exact formula

$$F_0(x) = \frac{1}{2} + \frac{1}{\pi}\left(\arcsin x + x\sqrt{1 - x^2}\right).$$

The maximum difference was approximately

$$2.03 \times 10^{-6}.$$

The Gaussian target CDF

$$F_1(y) = \Phi(y)$$

was also checked and found to be monotone.

Therefore, both CDFs were computed appropriately.

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3. Generalized Inverse and Quantile Function

The optimal map was computed using

$$T(x) = \Phi^{-1}(F_0(x)).$$

The Gaussian quantile function was evaluated using the percent-point function.

The LLM-generated code also included a separate numerical inverse-CDF procedure based on interpolation.

The interpolated inverse was compared with the Gaussian quantile function, and the maximum difference was approximately

$$1.33 \times 10^{-7}.$$

The code also verified the fundamental identity

$$\Phi(T(x)) = F_0(x).$$

The maximum error was

$$2.22 \times 10^{-16},$$

which is essentially machine precision.

Thus the LLM correctly used a quantile or generalized-inverse construction rather than assuming that an arbitrary CDF has an elementary inverse.

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4. Compact Support

The source density is supported only on

$$[-1, 1].$$

At the endpoints,

$$F_0(-1) = 0$$

and

$$F_0(1) = 1.$$

Therefore,

$$T(-1) = \Phi^{-1}(0) = -\infty$$

and

$$T(1) = \Phi^{-1}(1) = +\infty.$$

A numerical program cannot directly use these infinite values.

The LLM-generated code therefore excluded the exact endpoints from the particle grid.

This is a reasonable numerical choice, provided that it is understood as a computational approximation rather than a change in the mathematics.

The code separately examined points increasingly close to the endpoints.

For example,

$$T(0.9) \approx 2.0815,$$

$$T(0.99) \approx 3.2392,$$

$$T(0.9999) \approx 4.8556,$$

and

$$T(1 - 10^{-8}) \approx 7.1025.$$

Thus the numerical results correctly demonstrate

$$T(x) \to +\infty$$

as

$$x \to 1^{-}.$$

By symmetry,

$$T(x) \to -\infty$$

as

$$x \to -1^{+}.$$

Therefore, the compact support was handled correctly.

One minor weakness is that one final diagnostic line saying that endpoint divergence had been demonstrated was hard-coded as `True`.

That statement itself is not evidence. The actual endpoint calculations are what verify the claim.

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5. Gaussian Tails

The Gaussian target has support on all of

$$\mathbb{R}.$$

Therefore, no finite numerical grid can represent it exactly.

The LLM-generated code handled this issue honestly.

Rather than claiming that

$$[-6, 6]$$

contains the entire Gaussian, it explicitly calculated the probability mass outside the finite interval.

The truncation experiment gave approximately:

L	**Gaussian mass outside** $[-L, L]$
2	4.55×10^{-2}
3	2.70×10^{-3}
4	6.33×10^{-5}
5	5.73×10^{-7}
6	1.97×10^{-9}

The generated code also compared the transport maps obtained from the full Gaussian and from Gaussian distributions truncated to

$$[-L, L].$$

For the central probability region

$$0.001 \leq F_0(x) \leq 0.999,$$

the maximum transport-map errors were approximately

$$1.1068$$

for $L = 2$,

$$0.2625$$

for $L = 3$,

$$0.00923$$

for $L = 4$,

and

$$8.47 \times 10^{-5}$$

for $L = 5$.

This showed quantitatively how quickly the finite-domain approximation improves.

The LLM therefore did not make the mistake of pretending that infinity fits on a finite numerical grid.

Additional Verification of the Generated Code

The generated code also checked conservation of probability mass during the interpolation.

The numerical masses were approximately

$$0.99999963,$$

$$1.00000613,$$

$$1.00000615,$$

$$1.00000616,$$

and

$$1.00000616$$

for

$$t = 0,\ 0.25,\ 0.5,\ 0.75,\ 1.$$

The errors were therefore only on the order of

$$10^{-6}.$$

At

$$t = 1,$$

the interpolated density was compared directly with the standard Gaussian.

The maximum error was

$$5.55 \times 10^{-17}.$$

These additional checks support the correctness of the generated implementation.

Where the LLM Was Useful

The LLM was useful for:

- organizing the numerical optimal transport procedure;
- deriving the CDF and quantile construction;
- generating Python code for the transport map;
- generating the displacement interpolation;
- implementing the change-of-variables formula for the density;
- suggesting numerical consistency checks;
- constructing the Gaussian truncation experiment.

The generated code also included several useful diagnostics that made verification easier.

Where the LLM Could Be Mathematically Careless

Even though the code performed well, several aspects could have been misleading if they were accepted without inspection.

First, the program prints its own `PASS` statements.

A printed `PASS` statement does not independently prove that the test itself is mathematically valid.

Second, the code clips probability levels away from exactly

$$0$$

and

$$1$$

before applying the Gaussian quantile function.

This is necessary numerically, but it could hide the true mathematical endpoint divergence if the user did not examine the limiting behavior separately.

Third, finite-grid calculations only approximate the full continuous problem.

For example, mass conservation was accurate only to about

$$10^{-6},$$

not exact.

Fourth, a finite plot of the Gaussian cannot demonstrate that its support is all of

$\mathbb{R}$.

That conclusion comes from the mathematical form of the Gaussian density, not from the plot.

Finally, one endpoint-divergence diagnostic was hard-coded rather than computed automatically.

Therefore, the generated program still required mathematical inspection.

Overall LLM Critique

For this homework, the LLM-generated reasoning and code were largely correct.

It correctly handled:

- probability normalization;
- monotone CDFs;
- the quantile function;
- numerical generalized inversion;
- compact source support;
- endpoint divergence;
- displacement interpolation;
- probability conservation;
- Gaussian tails;
- finite Gaussian truncation.

The strongest numerical checks were

$$\max |\Phi(T(x)) - F_0(x)| \approx 2.22 \times 10^{-16}$$

and

$$\max |\rho_{t=1}(y) - \rho_1(y)| \approx 5.55 \times 10^{-17}.$$

However, the experiment also demonstrates why LLM-generated mathematical code should not be trusted solely because it runs successfully.

The formulas, limiting behavior, numerical invariants, and error sizes must still be checked independently.

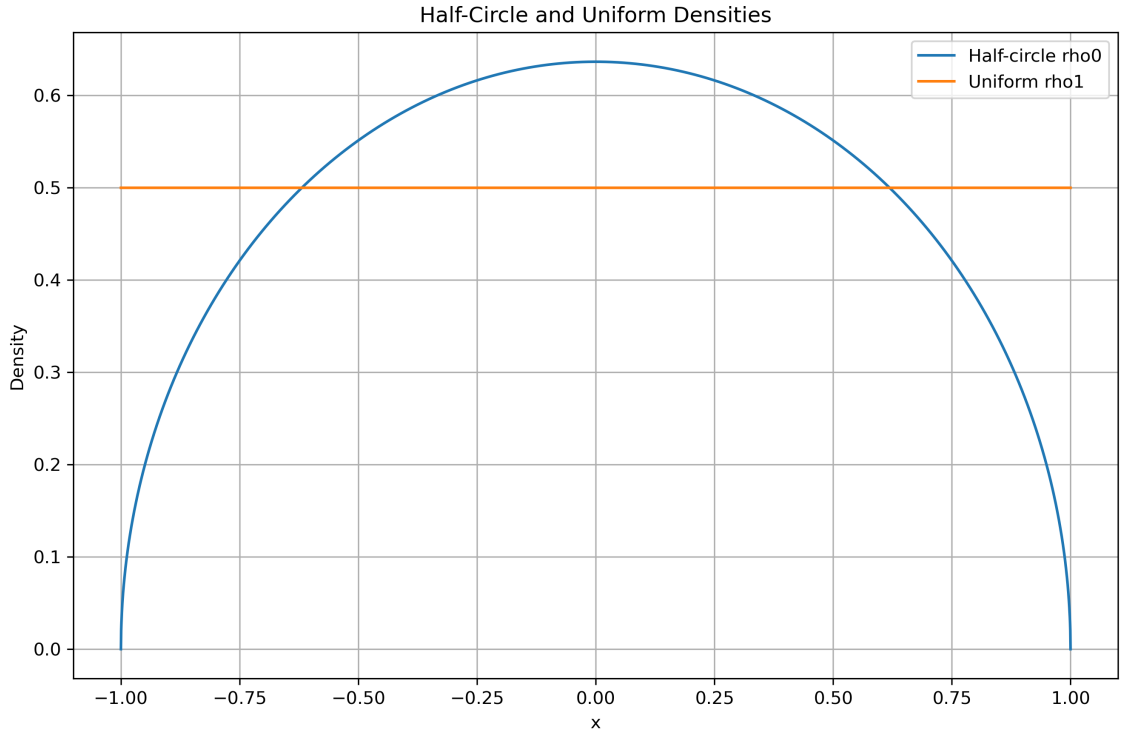
The LLM was therefore useful as a mathematical and computational assistant, but not as an unquestioned source of truth.

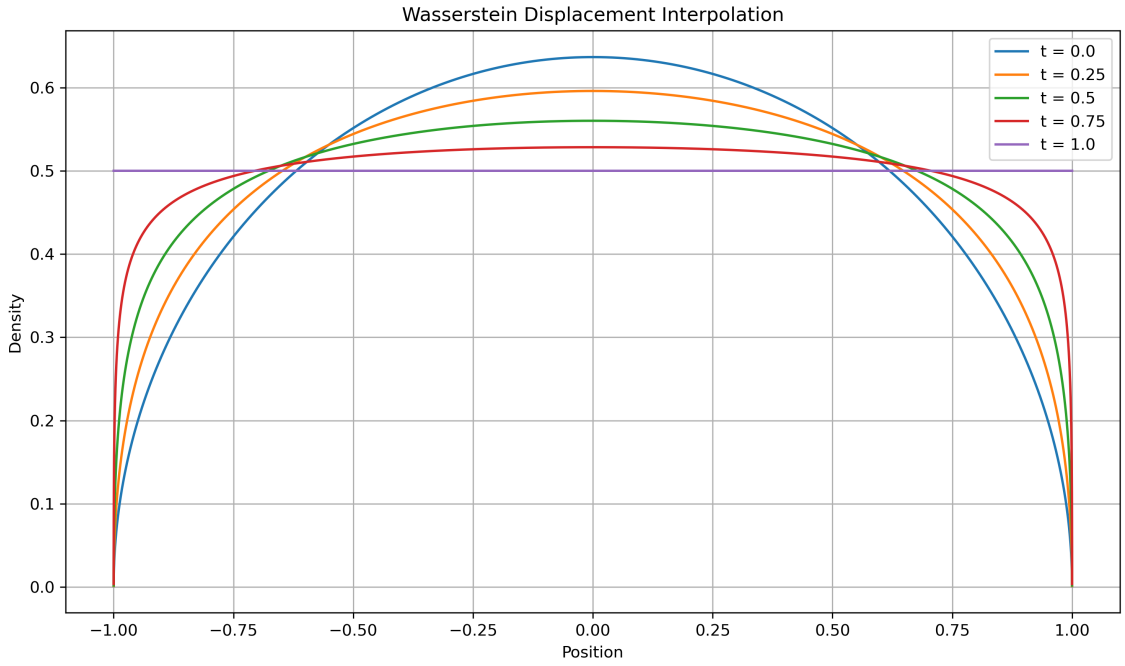
Deliverables Summary

All six requested deliverables are present: the source/target/interpolated-density plots, both optimal-map plots, the numerical value of W_2^2, the numerical Benamou-Brenier action, the compact-support/Gaussian-tail discussion, and the LLM prompts with code critique.

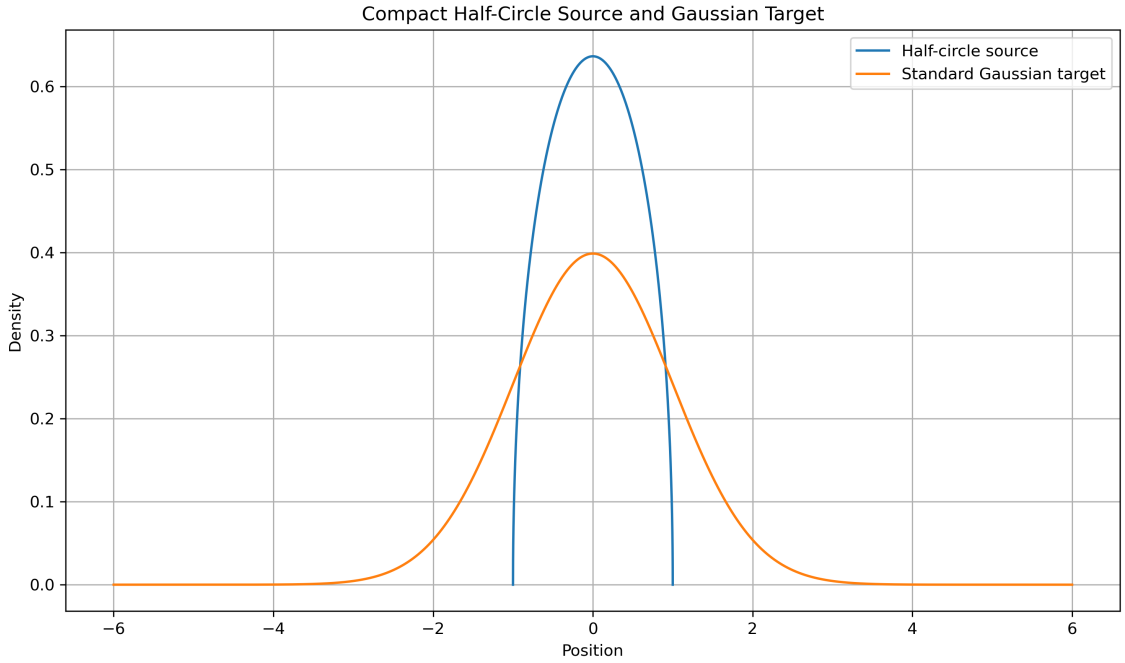
1. Plots of ρ_0, ρ_1, and ρ_t

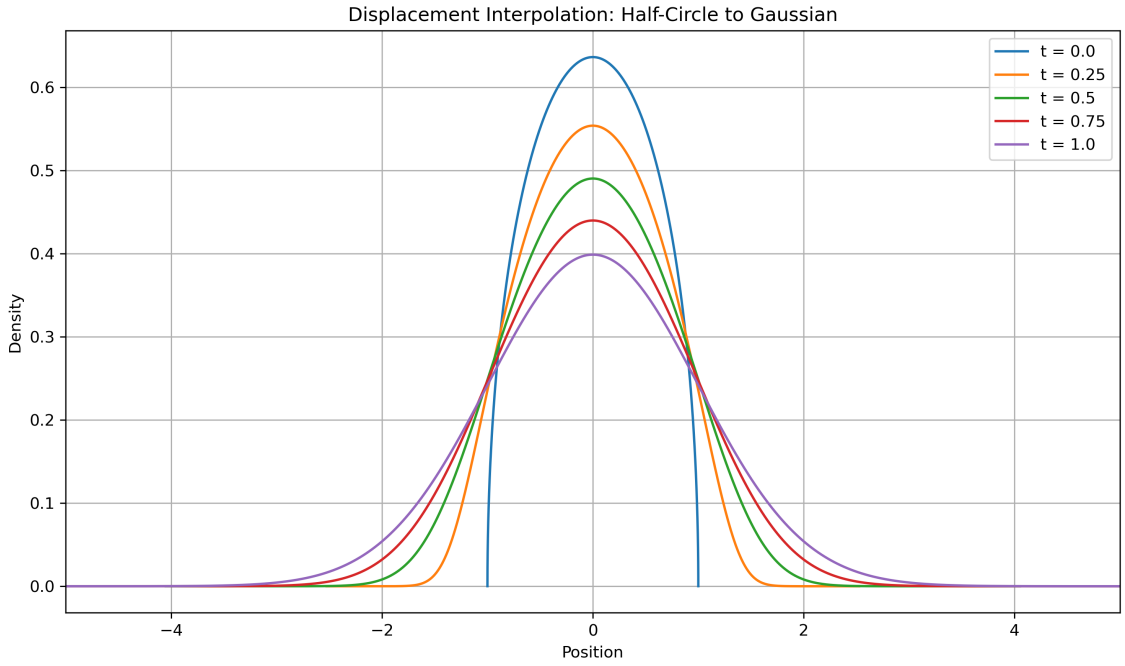
For the half-circle to uniform experiment:





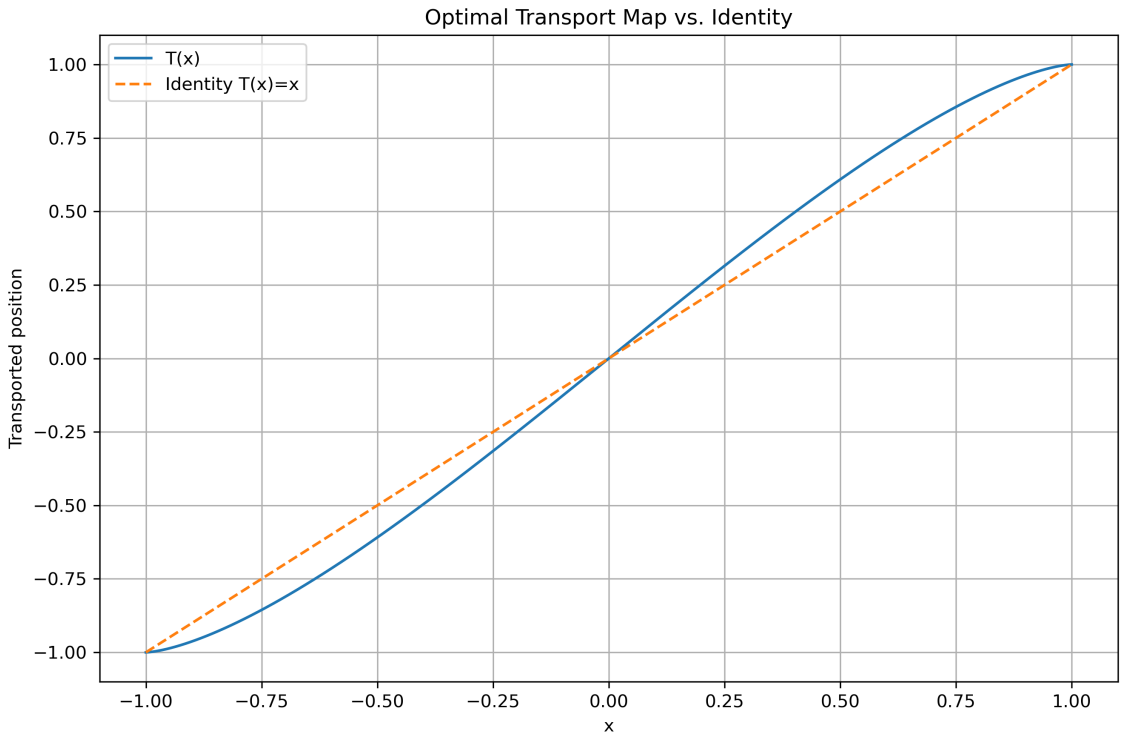
For the half-circle to Gaussian experiment:



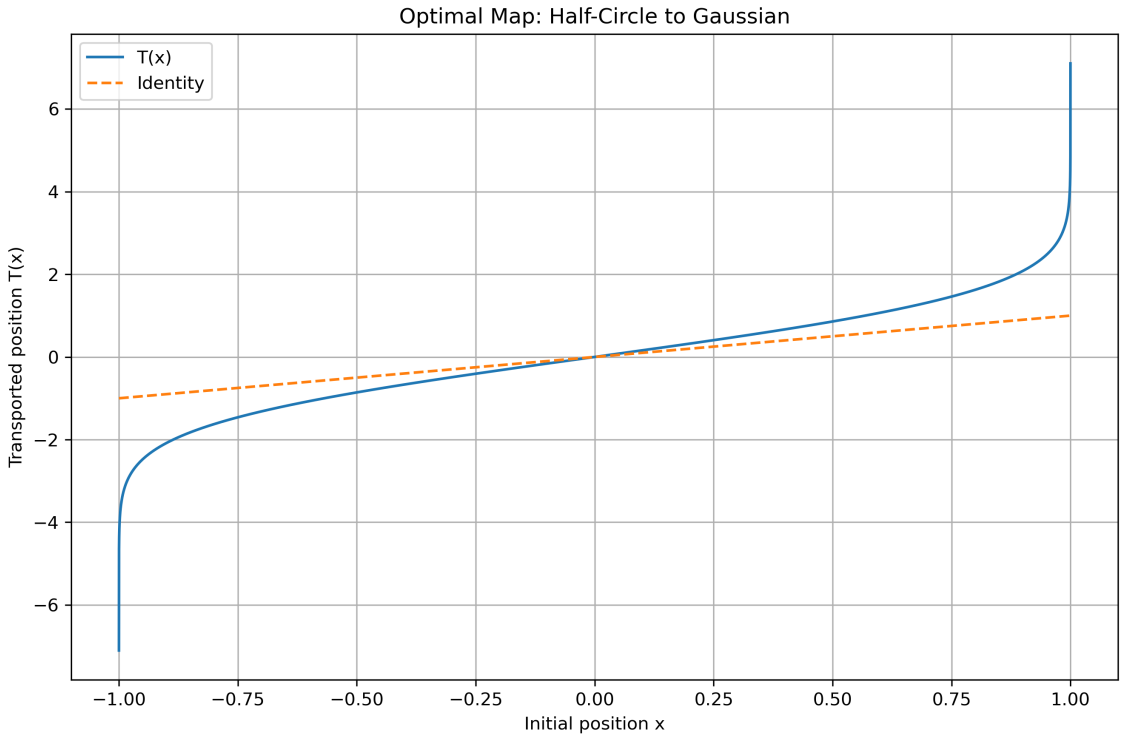


2. Plots of the Optimal Maps $T(x)$

For the uniform target:



For the Gaussian target:



3. Numerical Estimate of W_2^2

For the half-circle to uniform experiment,

$$W_2^2 \approx 0.00692838.$$

4. Numerical Estimate of the Benamou-Brenier Action

The dynamic calculation gave

$$\mathcal{A} \approx 0.00692838.$$

Therefore,

$$\mathcal{A} \approx W_2^2.$$

The difference was only on the scale of approximately

$$10^{-9}.$$

5. Compact Support Versus Gaussian Tails

The uniform target remains supported on the same compact interval

$$[-1, 1].$$

The Gaussian target instead has support on

$$\mathbb{R}.$$

Consequently, the Gaussian transport map must satisfy

$$T(x) \to -\infty$$

as

$$x \to -1^{+}$$

and

$$T(x) \to +\infty$$

as

$$x \to 1^{-}.$$

This causes the displacement interpolation to develop tails extending over all of

$$\mathbb{R}$$

for every positive time.

Numerically, truncating the Gaussian at approximately

$$L = 4$$

is visually reasonable, while

$$L = 5$$

is highly accurate.

6. LLM Prompts and Critique

The LLM was first used as a tutor to explain the CDF and quantile construction.

It was then asked to generate standalone Python code for the complete compact-support-to-Gaussian experiment.

The generated reasoning and code were checked independently for:

- normalization;
- CDF monotonicity;
- generalized inverse handling;
- compact-support behavior;
- Gaussian-tail treatment.

The generated solution performed well on all five criteria, but the exercise demonstrated that LLM output should still be verified rather than accepted solely because the code runs or prints successful diagnostic messages.

Final Conclusion

The experiments numerically demonstrate the one-dimensional Benamou-Brenier picture of optimal transport.

For the half-circle to uniform problem, the optimal transport map is

$$T(x) = \frac{2}{\pi}\left(\arcsin x + x\sqrt{1-x^2}\right).$$

The displacement interpolation continuously spreads probability mass away from the center until the density becomes uniform.

The squared Wasserstein distance was approximately

$$W_2^2 \approx 0.00692838,$$

while the numerical Benamou-Brenier action was also approximately

$$0.00692838.$$

Their agreement numerically verifies

$$W_2^2(\rho_0,\rho_1) = \int_0^1 \int_{\mathbb{R}} |v_t(y)|^2 \rho_t(y)$$
$$, dy$$
$$, dt$$

along the Wasserstein geodesic.

When the target was replaced by a Gaussian, the optimal transport map became unbounded near the endpoints of the compact source interval.

This illustrates how optimal transport can move a compactly supported distribution into one with support over the entire real line.

Overall, the analytic formulas, numerical transport maps, interpolated densities, Wasserstein calculation, Benamou-Brenier action, mass-conservation checks, and Gaussian truncation study all agreed with the theoretical predictions.